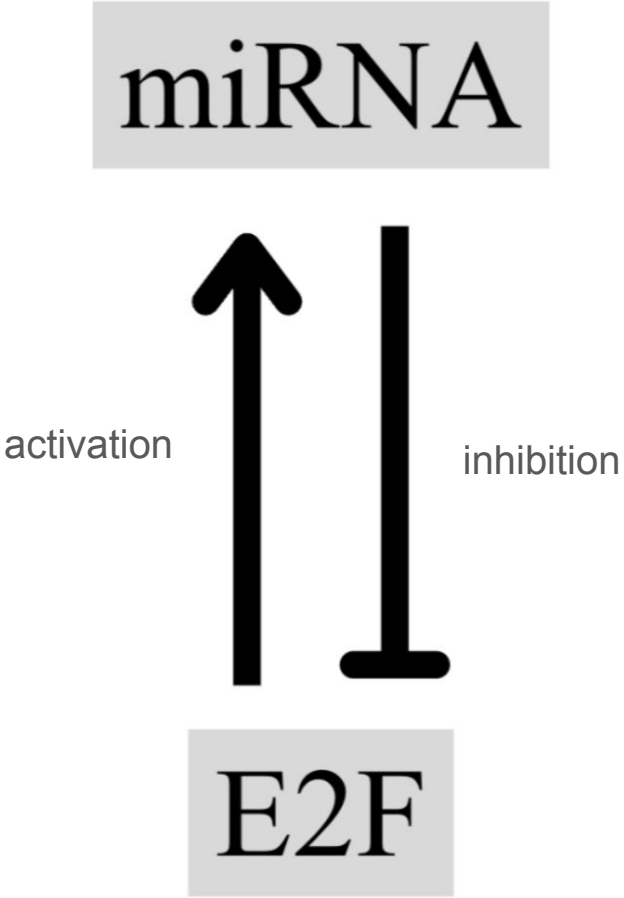
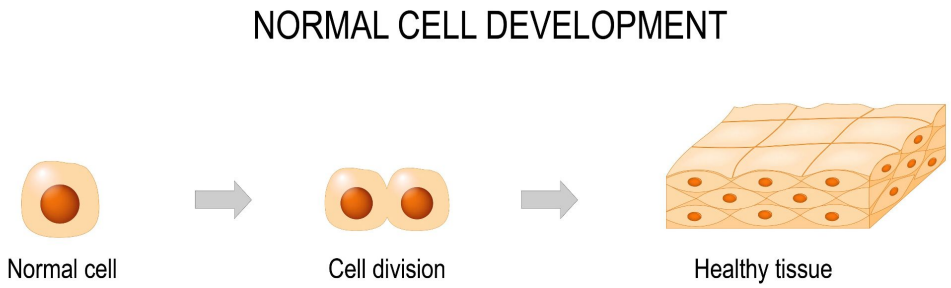
Emergence of bistability in a time delayed stochastic miRNA-protein toggle switch

AM 4999Z: March 27th, 2026

Background Information


miRNA
activation
inhibition
E2F

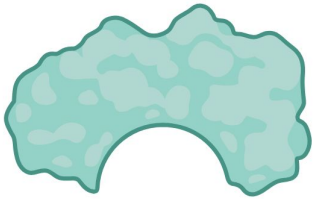

NORMAL CELL DEVELOPMENT
Normal cell
Cell division
Healthy tissue

Previous Modelling Efforts

$$\dot{p} = a + \frac{k_1 p^2}{G_1 + p^2 + G_2 m} - dp$$

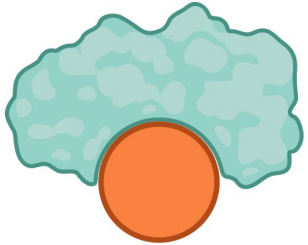
$$\dot{m} = b + k_2 p - \gamma m$$

protein

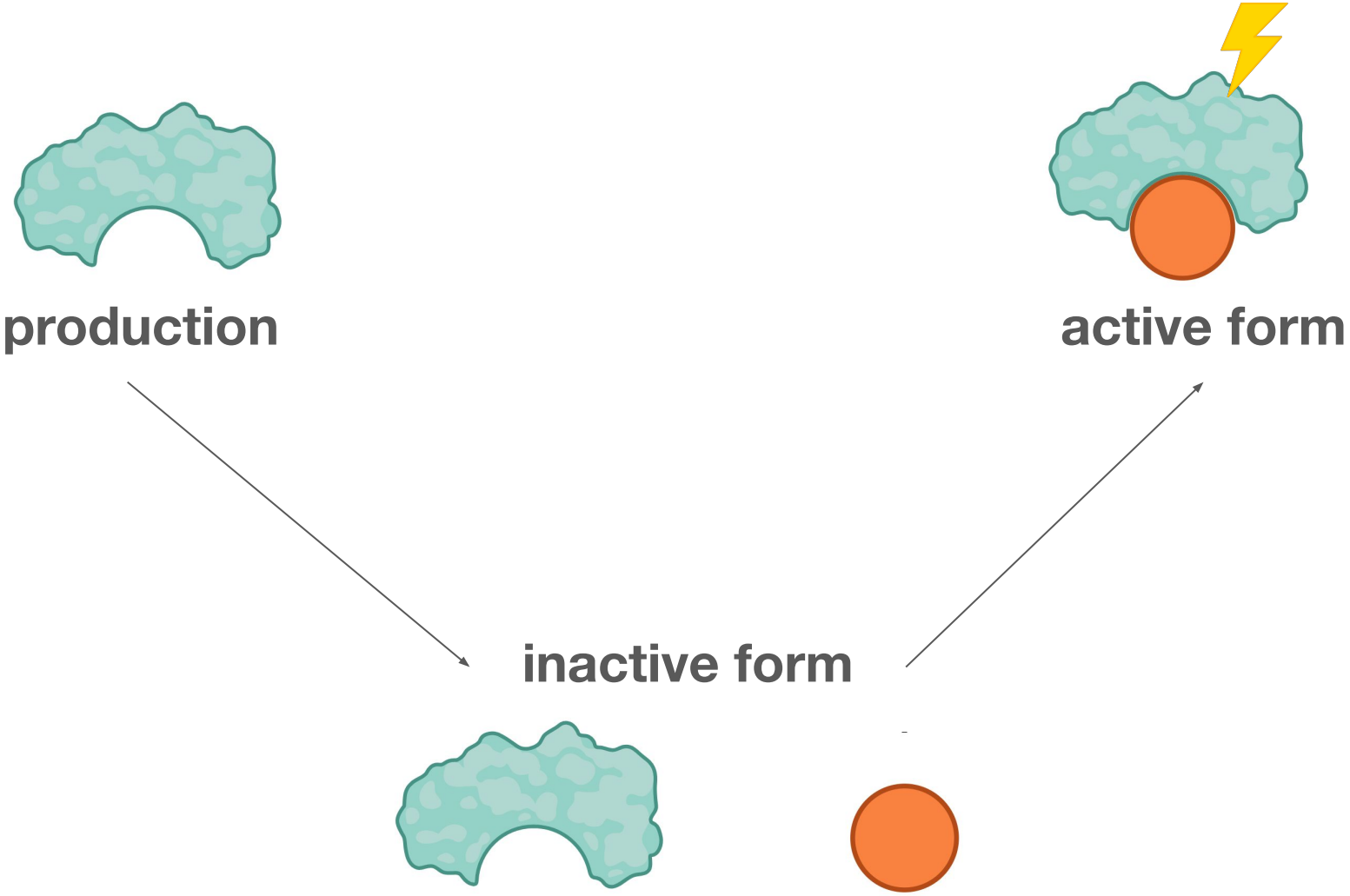


production

protein instantly
binding with miRNA



active form

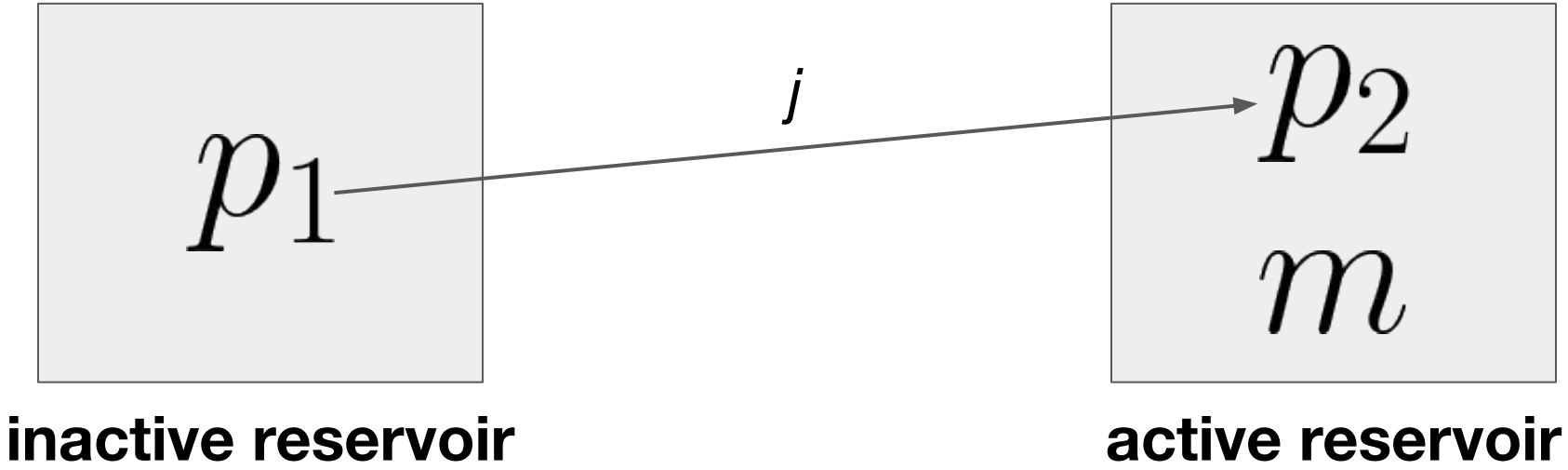

production
active form
inactive form

Our Modified Model

$$\dot{p_1} = a - dp_1 - jp_1$$

$$\dot{p_2} = \frac{k_1(p_2)^2}{G_1 + (p_2)^2 + G_2 m} - dp_2 + jp_1$$

$$\dot{m} = b + k_2 p_2 - \gamma m$$


p_1
j
p_2
m
inactive reservoir
active reservoir

Implementation via Gillespie’s Algorithm

unmodified system at time *t*

p **production**

p **degradation**

p **activation**

m **production**

m **degradation**

m **interaction with** p

unmodified system at time *t*

p **production**

$$p \xrightarrow{a} p+1$$

m **production**

$$m \xrightarrow{b} m+1$$

p **degradation**

$$p \xrightarrow{dp} p-1$$

m **degradation**

$$m \xrightarrow{\gamma m} m-1$$

p **activation**

$$p \xrightarrow{\frac{k_1 p^2}{G_1 + p^2 + G_2 m}} p+1$$

m **interaction with** p

$$m \xrightarrow{k_2 p} m+1$$

modified system at time *t*

p_1	production	m	production
p_1	degradation	m	degradation
p_2	activation	m	production from p_2
p_2	degradation	p_2	diffusion from p_1

modified system at time *t*

p_1 **production**

$$p_1 \xrightarrow{a} p_1 + 1$$

p_1 **degradation**

$$p_1 \xrightarrow{dp_1} p_1 - 1$$

p_2 **activation**

$$p_2 \xrightarrow{\frac{k_1(p_2)^2}{G_1 + (p_2)^2 + G_2 m}} p_2 + 1$$

p_2 **degradation**

$$p_2 \xrightarrow{dp_2} p_2 - 1$$

m **production**

$$m \xrightarrow{b} m + 1$$

m **degradation**

$$m \xrightarrow{\gamma m} m - 1$$

m **production from** p_2

$$m \xrightarrow{k_2 p_2} m + 1$$

p_2 **diffusion from** p_1

$$p_2 \xrightarrow{jp_1} p_2 + 1$$

$$p_1 \xrightarrow{jp_1} p_1 - 1$$

$$\lambda = \sum_{i=1}^{8} r_i$$

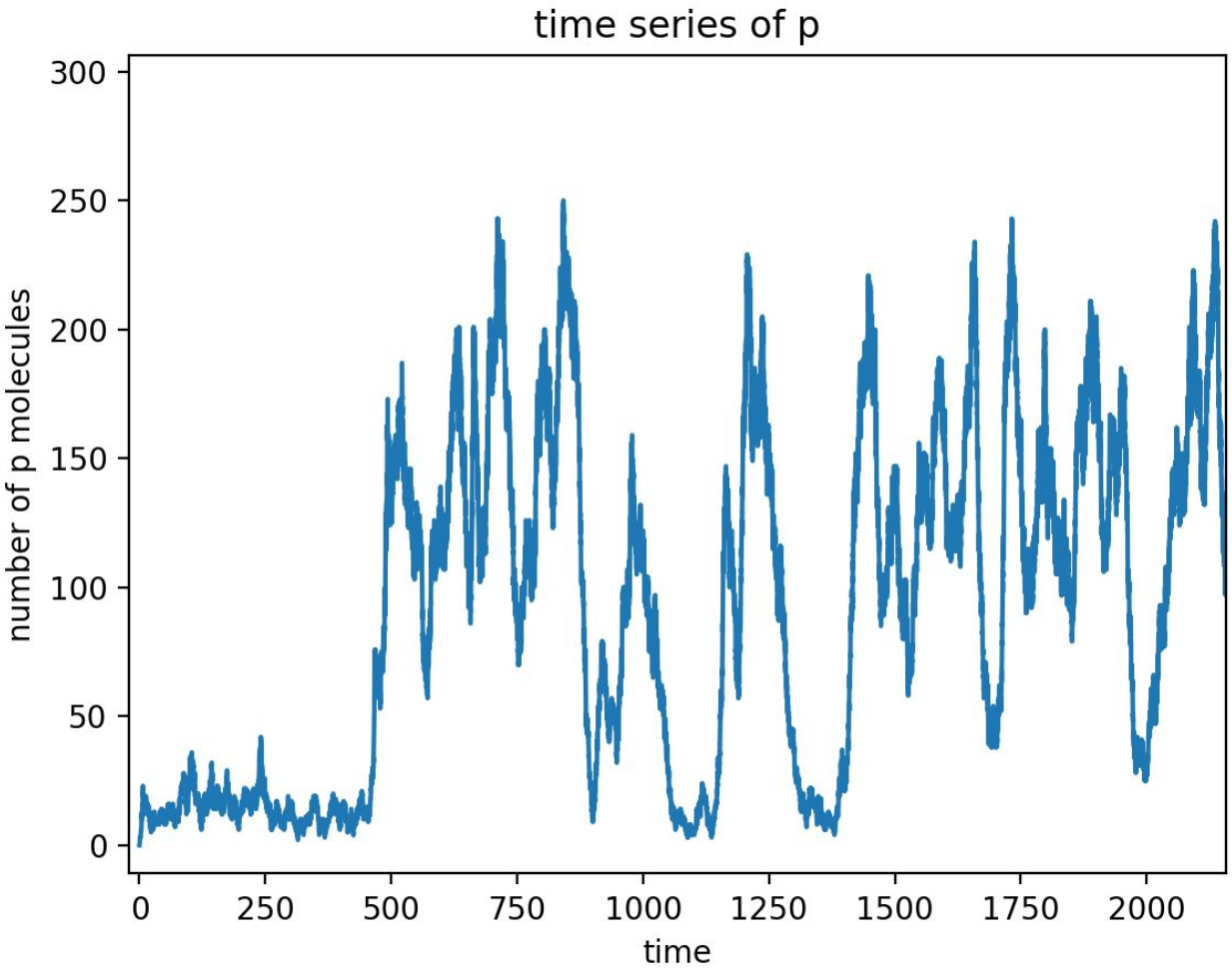
$$\tau = \frac{\textbf{sampling from exponential distribution}}{\lambda}$$

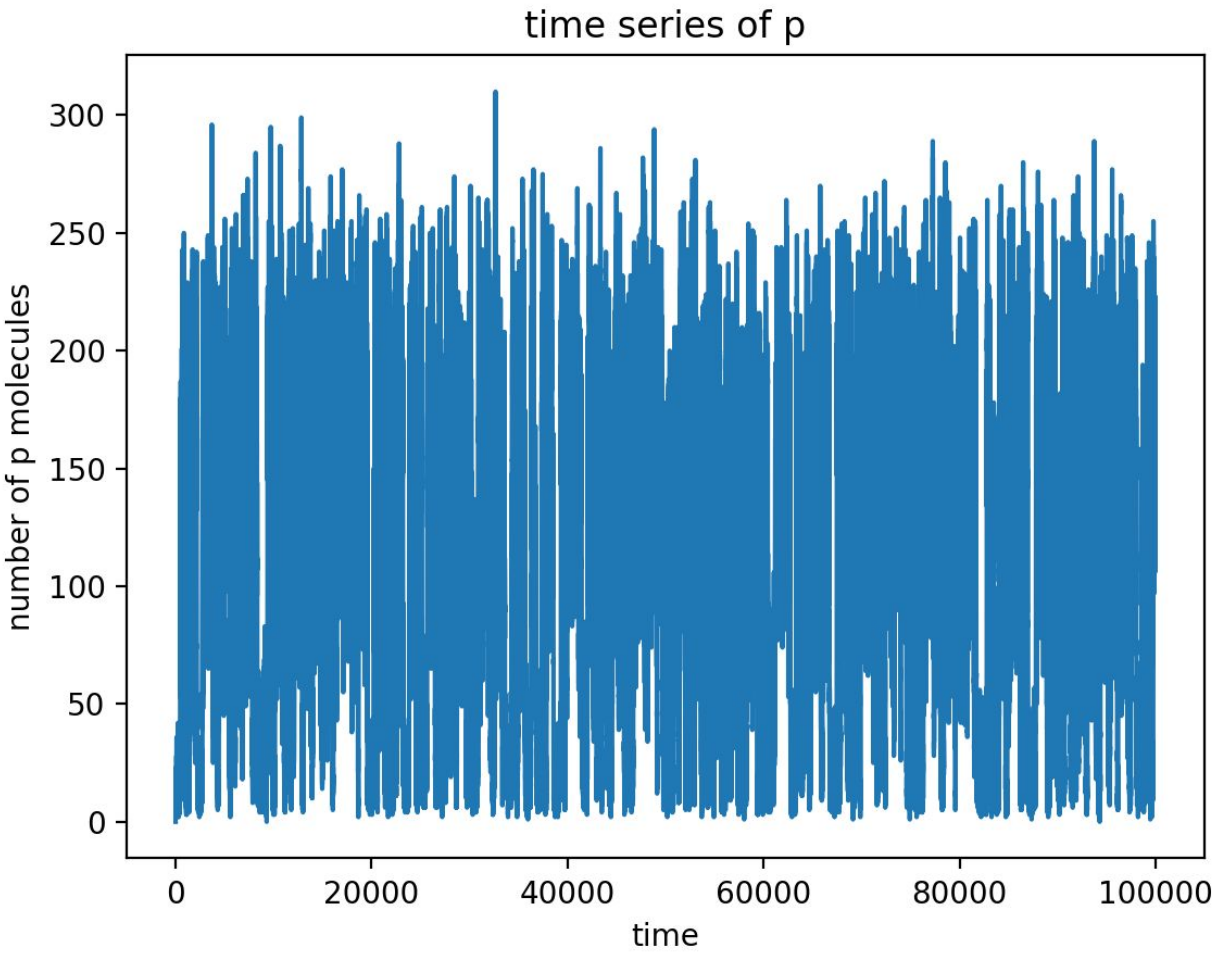
$$\frac{r_i}{\lambda}$$ = **frequency of the reaction**

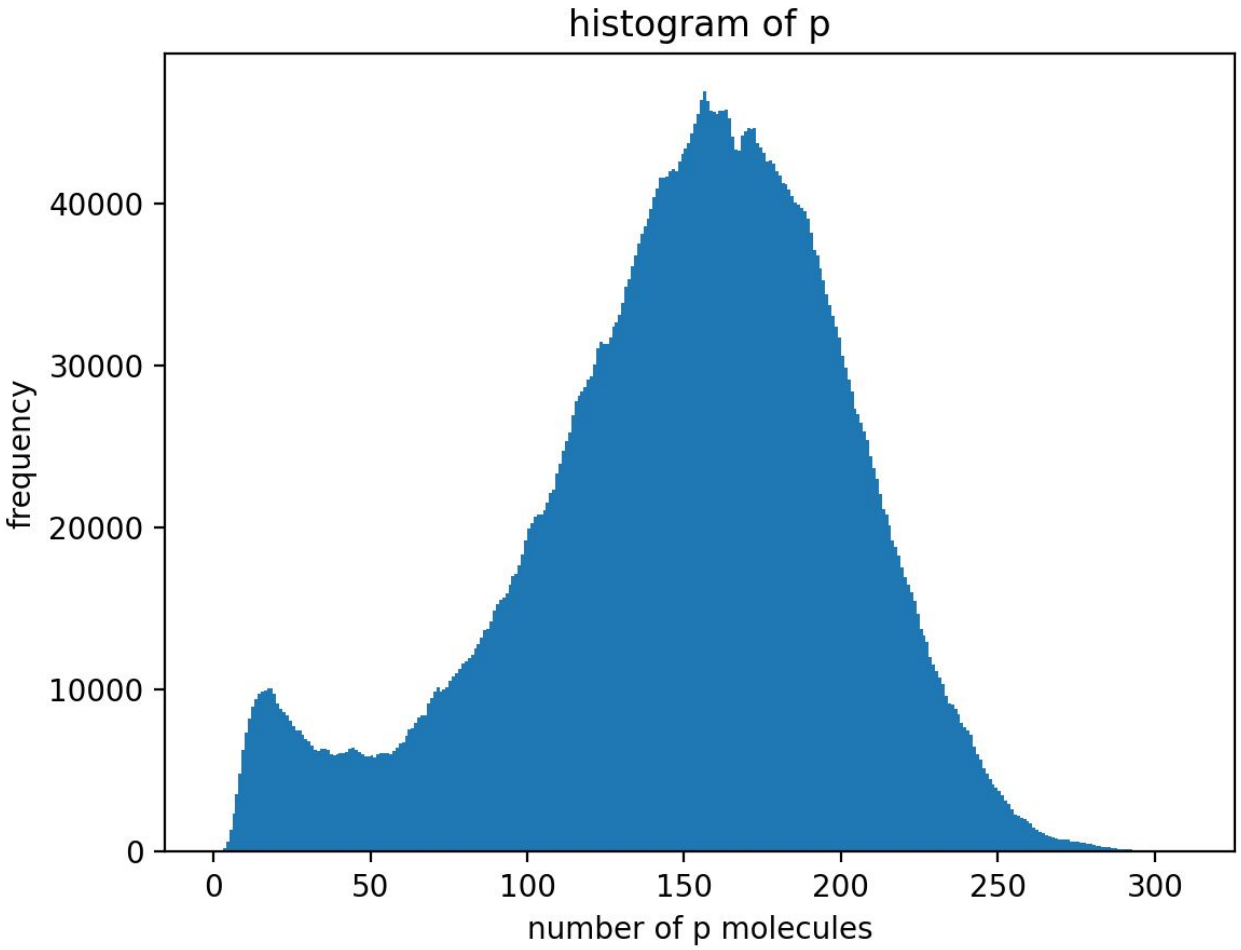
Results: Unmodified System

$$\dot{p} = a + \frac{k_1 p^2}{G_1 + p^2 + G_2 m} - dp$$

$$\dot{m} = b + k_2 p - \gamma m$$


time series of p
number of p molecules
time
0
50
100
150
200
250
300
0
250
500
750
1000
1250
1500
1750
2000


time series of p
number of p molecules
300
250
200
150
100
50
0
0
20000
40000
60000
80000
100000
time

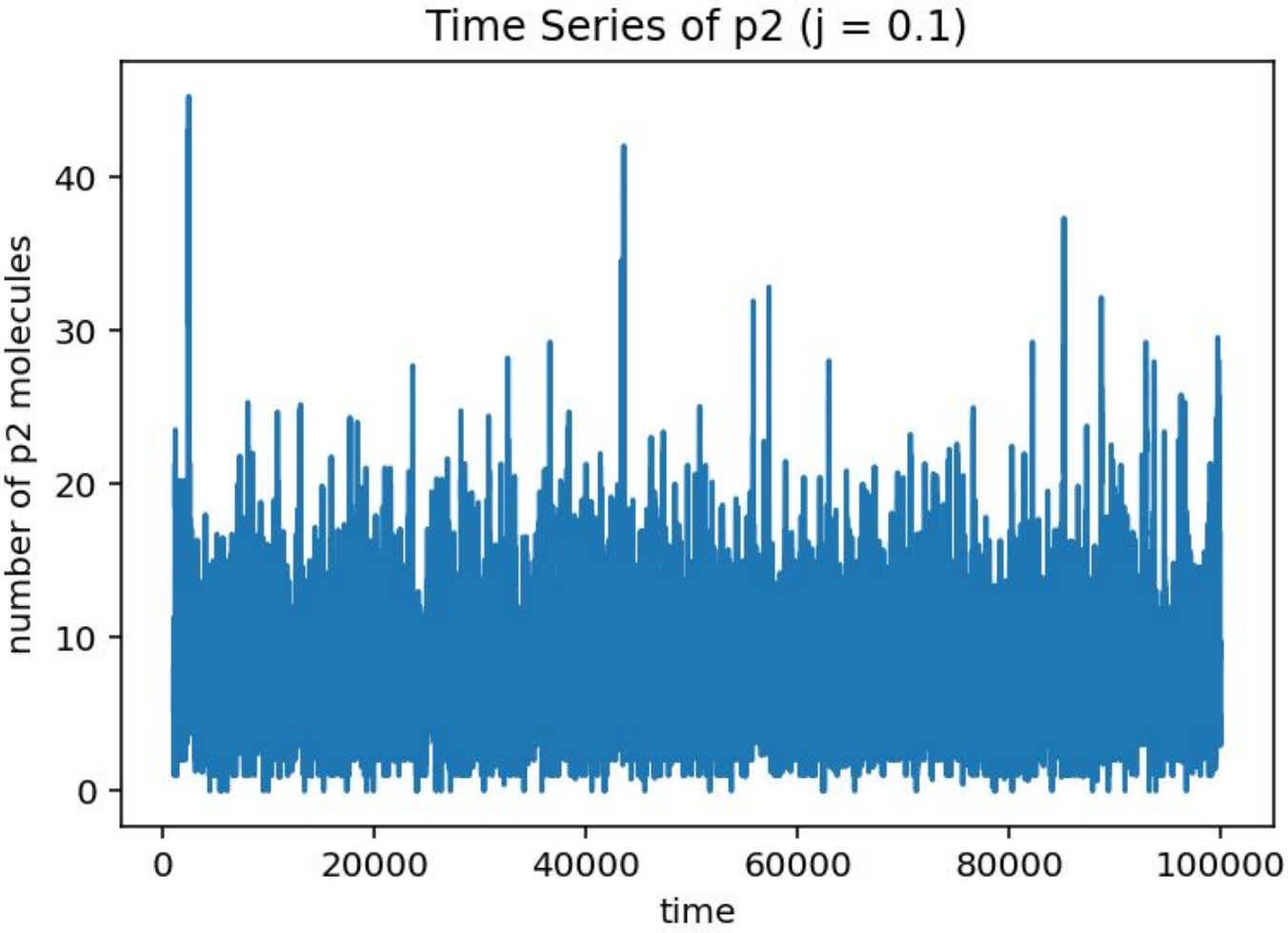

histogram of p
frequency
40000
30000
20000
10000
0
0
50
100
150
200
250
300
number of p molecules

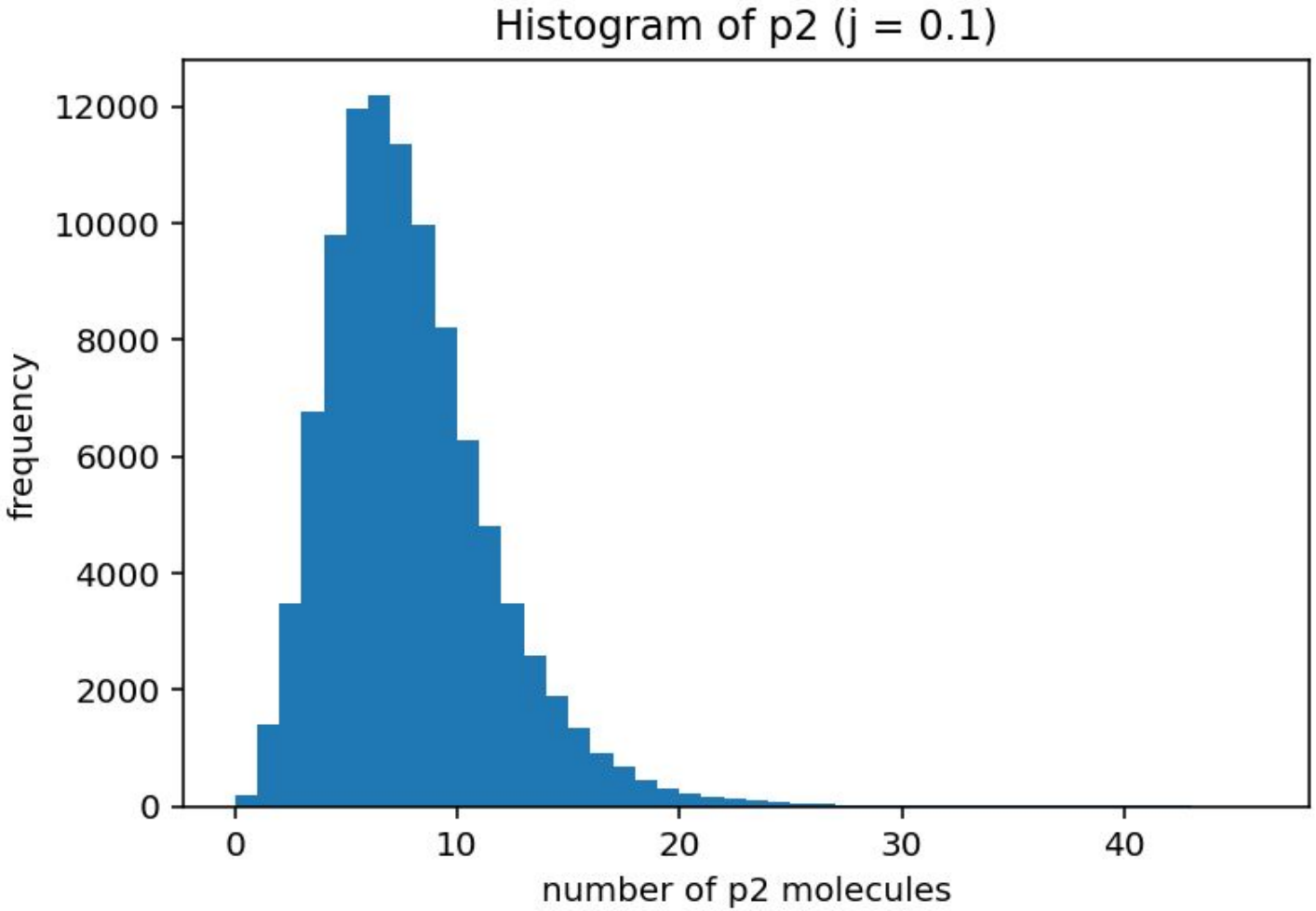
Results: Modified System

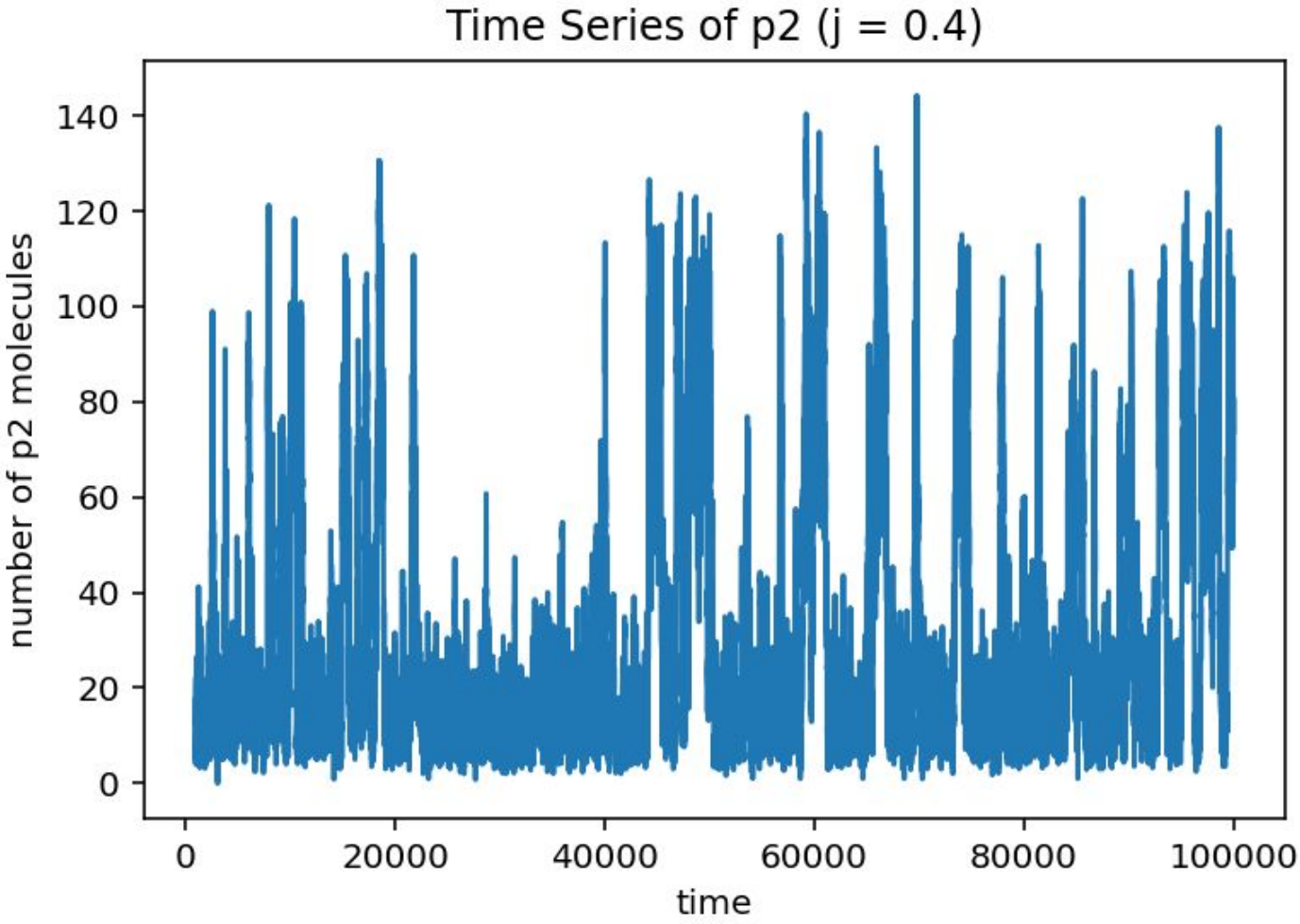
$$\dot{p_1} = a - dp_1 - jp_1$$

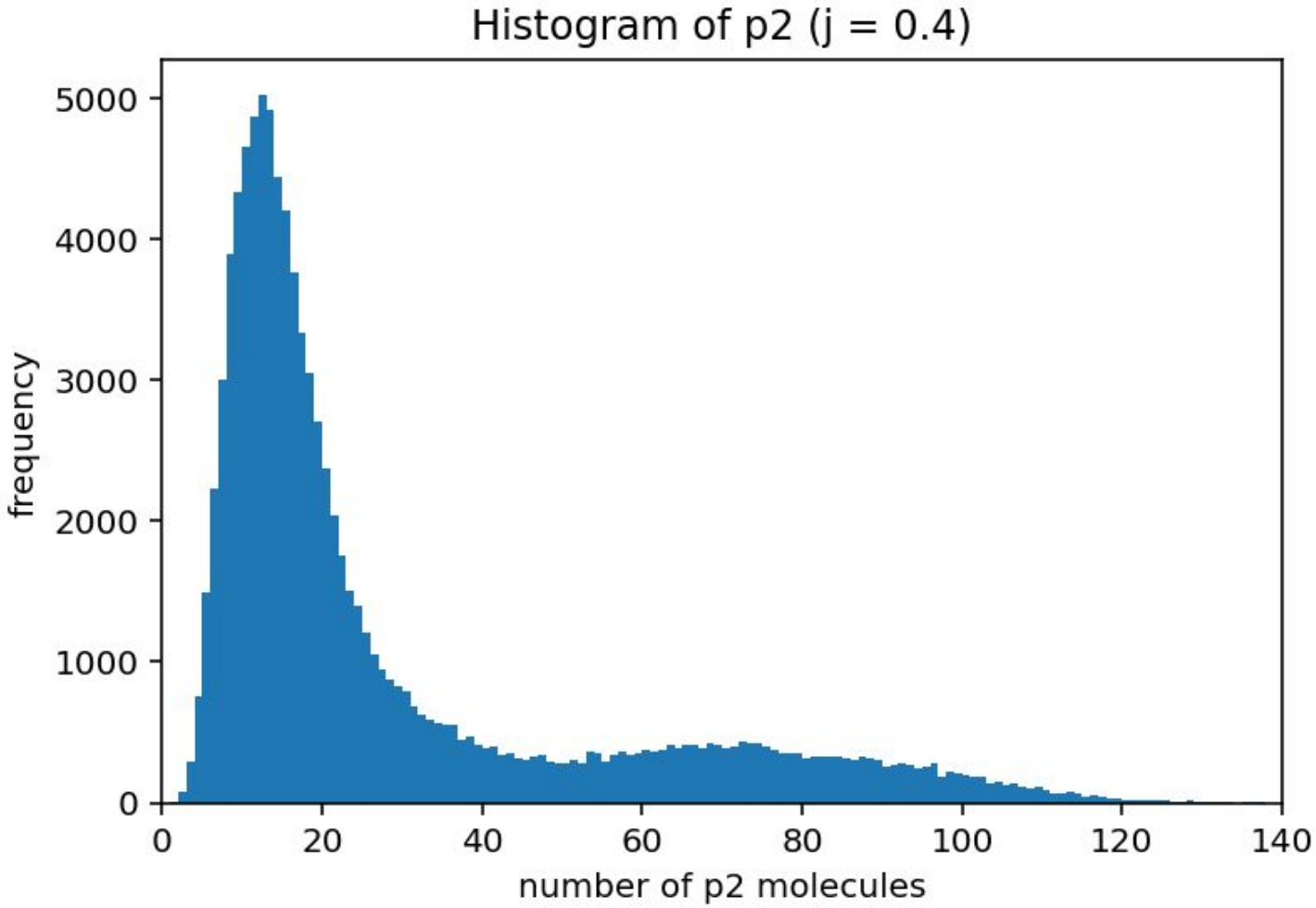
$$\dot{p_2} = \frac{k_1(p_2)^2}{G_1 + (p_2)^2 + G_2 m} - dp_2 + jp_1$$

$$\dot{m} = b + k_2 p_2 - \gamma m$$


Time Series of p2 (j = 0.1)
number of p2 molecules
0
10
20
30
40
0
20000
40000
60000
80000
100000
time

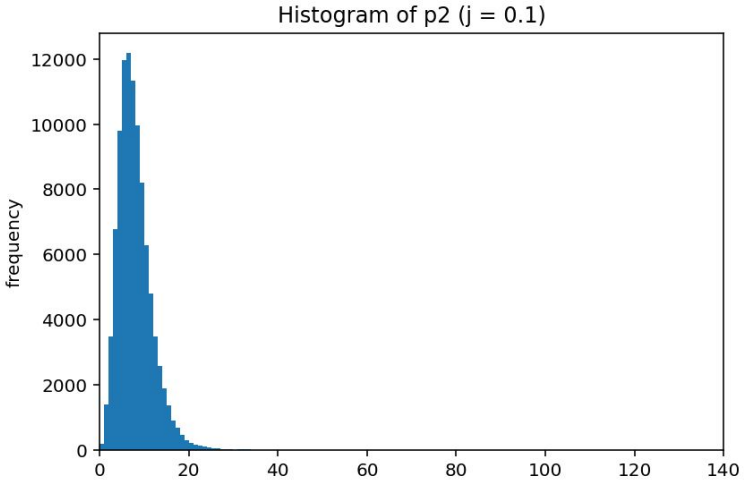

Histogram of p2 (j = 0.1)
frequency
12000
10000
8000
6000
4000
2000
0
0
10
20
30
40
number of p2 molecules

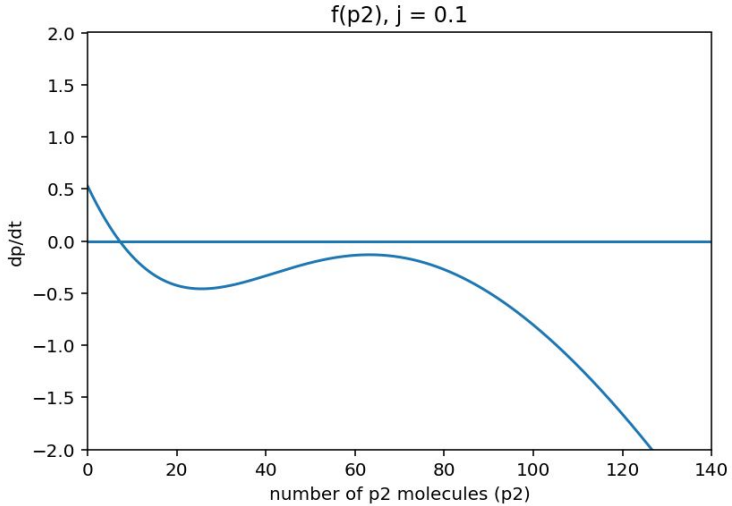

Time Series of p2 (j = 0.4)
number of p2 molecules
time
0
20
40
60
80
100
120
140
0
20000
40000
60000
80000
100000

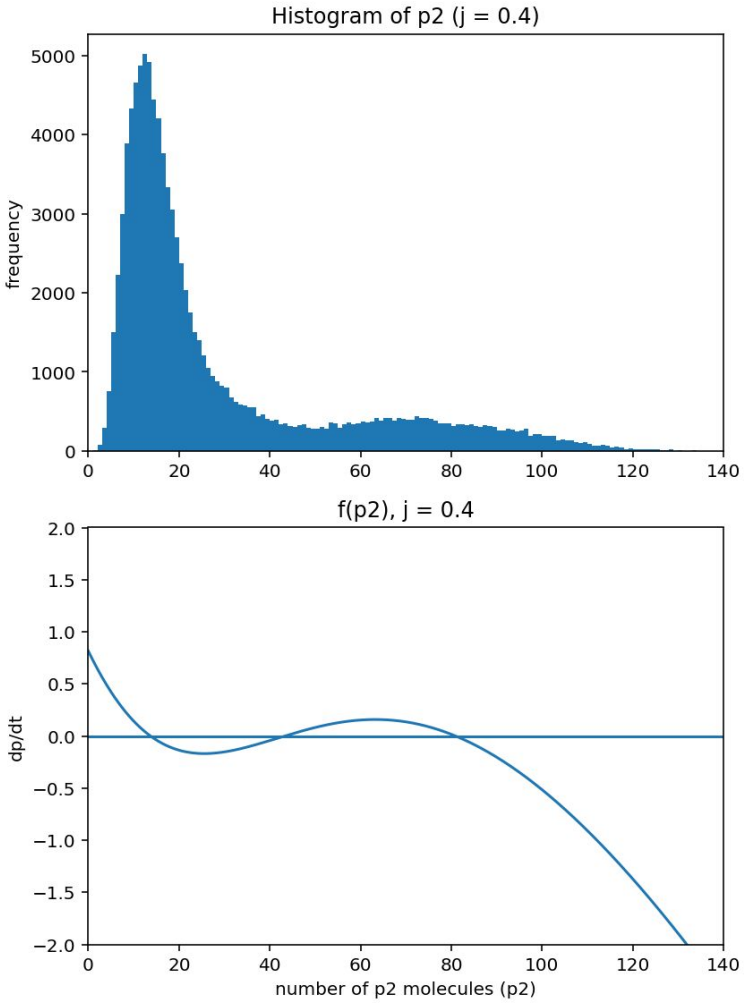

Histogram of p2 (j = 0.4)
frequency
number of p2 molecules
0
1000
2000
3000
4000
5000
0
20
40
60
80
100
120
140

Results:
Stability Analysis of Modified System

$$\frac{dp_2}{dt} = \frac{k_1(p_2^2)}{G_5 + p_2^2 + G_4 p_2} - dp_2 + j\left(\frac{a}{d+j}\right)$$


Histogram of p2 (j = 0.1)
frequency
12000
10000
8000
6000
4000
2000
0
0
20
40
60
80
100
120
140


f(p2), j = 0.1
dp/dt
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0
20
40
60
80
100
120
140
number of p2 molecules (p2)

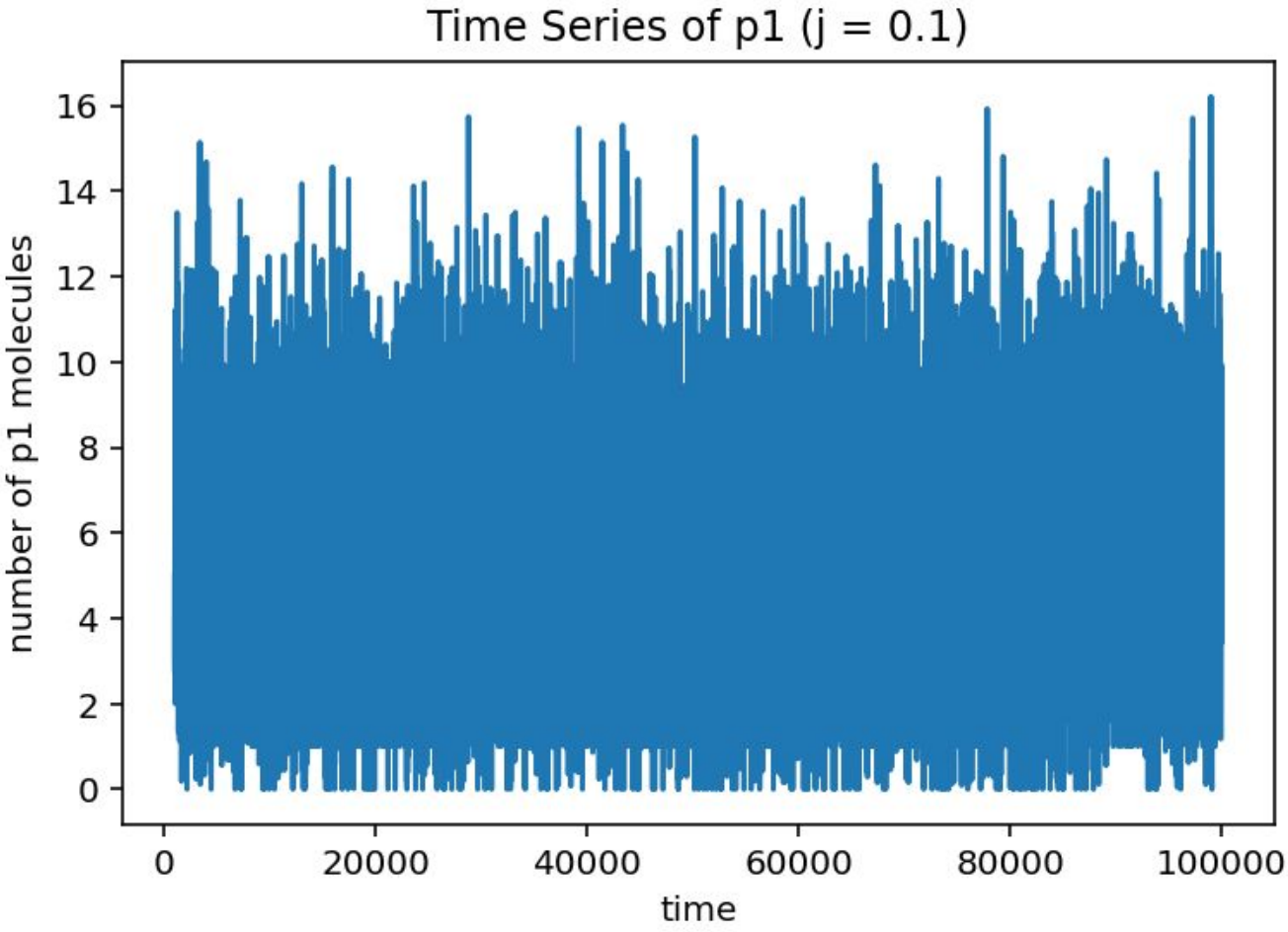

Histogram of p2 (j = 0.4)
frequency
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
f(p2), j = 0.4
dp/dt
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0
20
40
60
80
100
120
140
number of p2 molecules (p2)

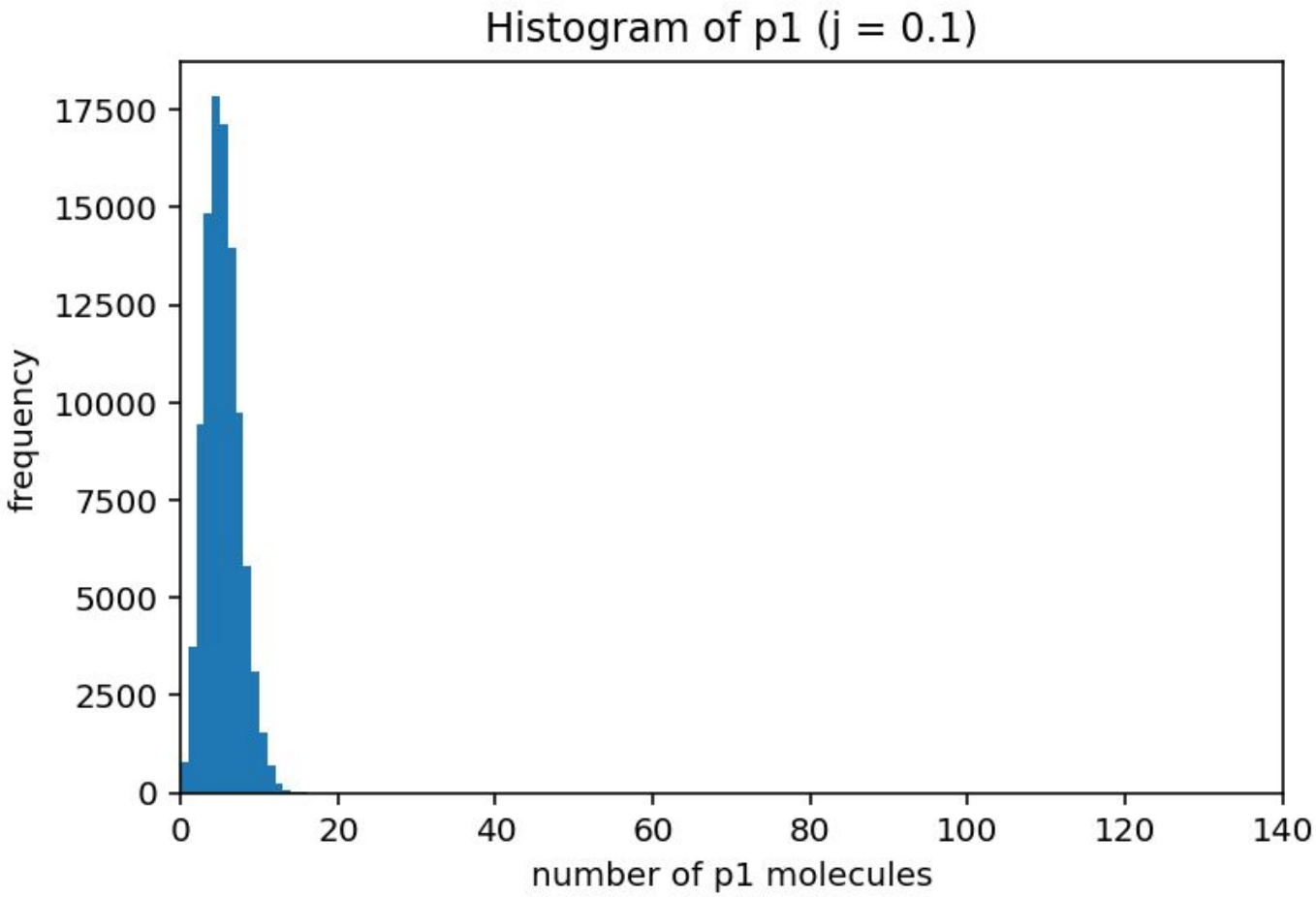
Discussion & Conclusion

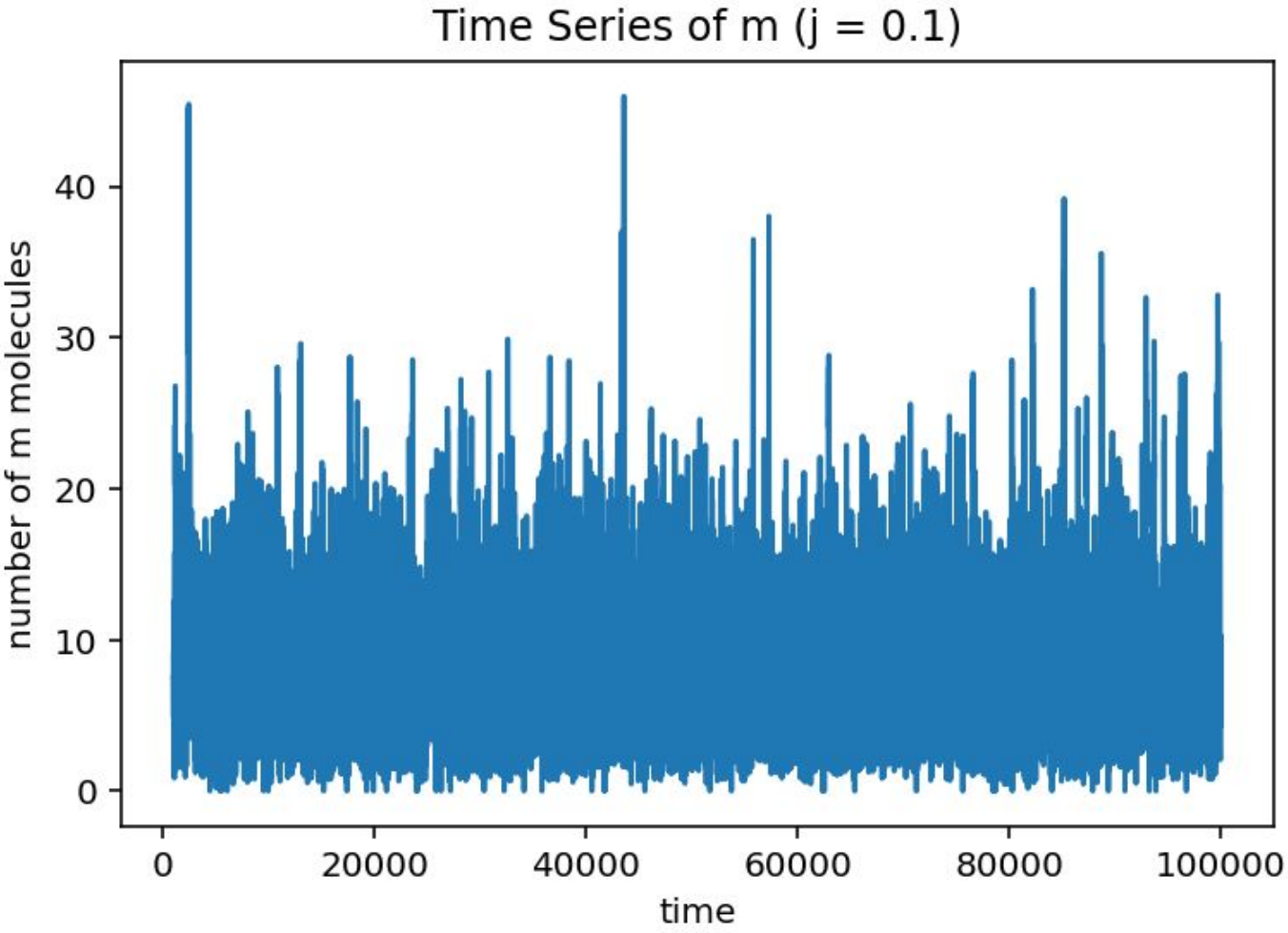
- Increasing j increases the conversion jp_1, raising the availability of active protein p_2
- Low j: system remains in one dominant, low p_2 range
- Higher j: system occupies both lower and higher p_2 ranges
- Stability analysis explains this by showing how the system is driven towards different p_2 levels
- Agreement between stochastic simulations and stability analysis supports this behaviour

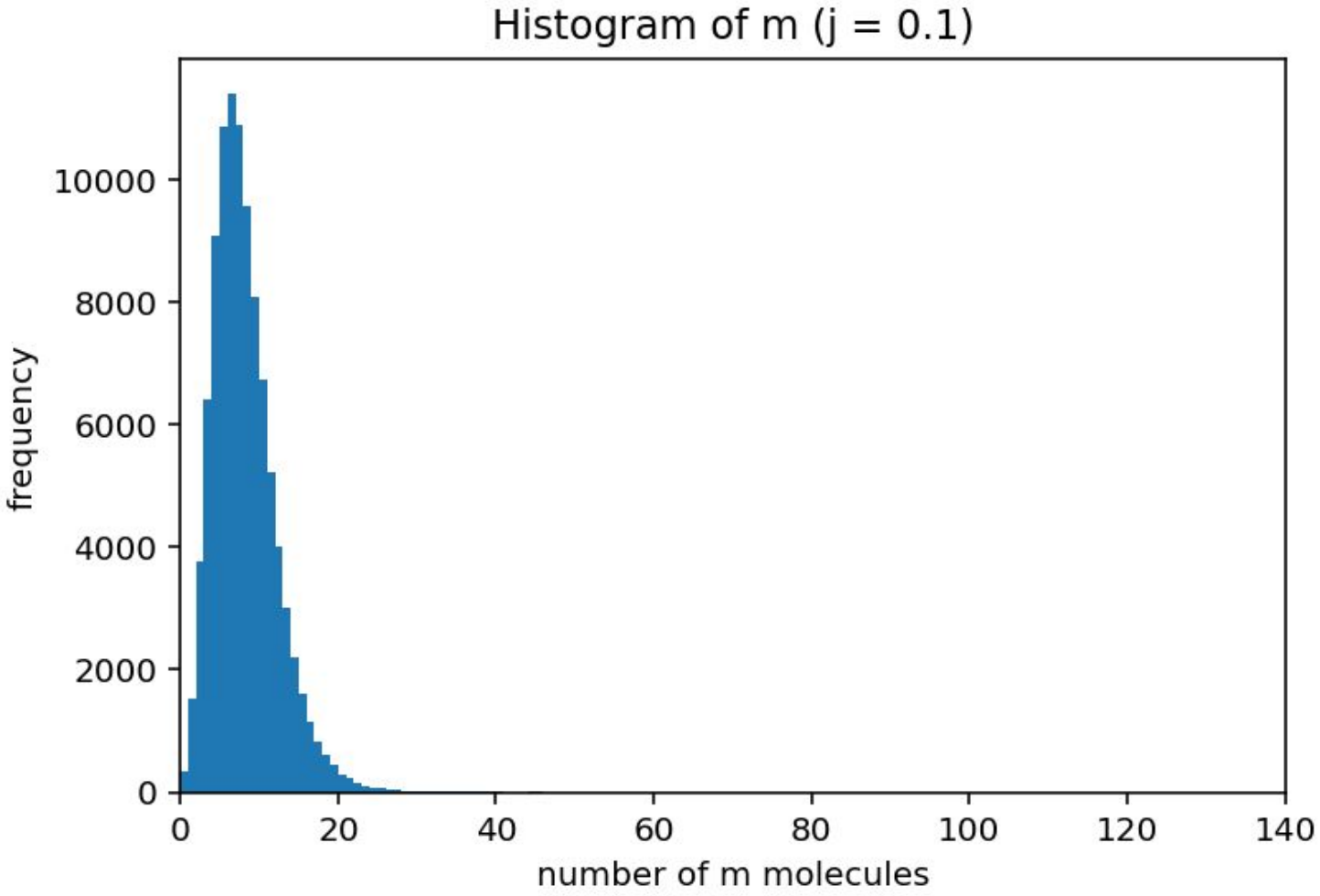
$$a = 10, k_1 = 12.5, G_1 = 72.8^2, G_2 = 10, \boxed{d = 0.09}, b = 0, k_2 = 10, \gamma = 10$$

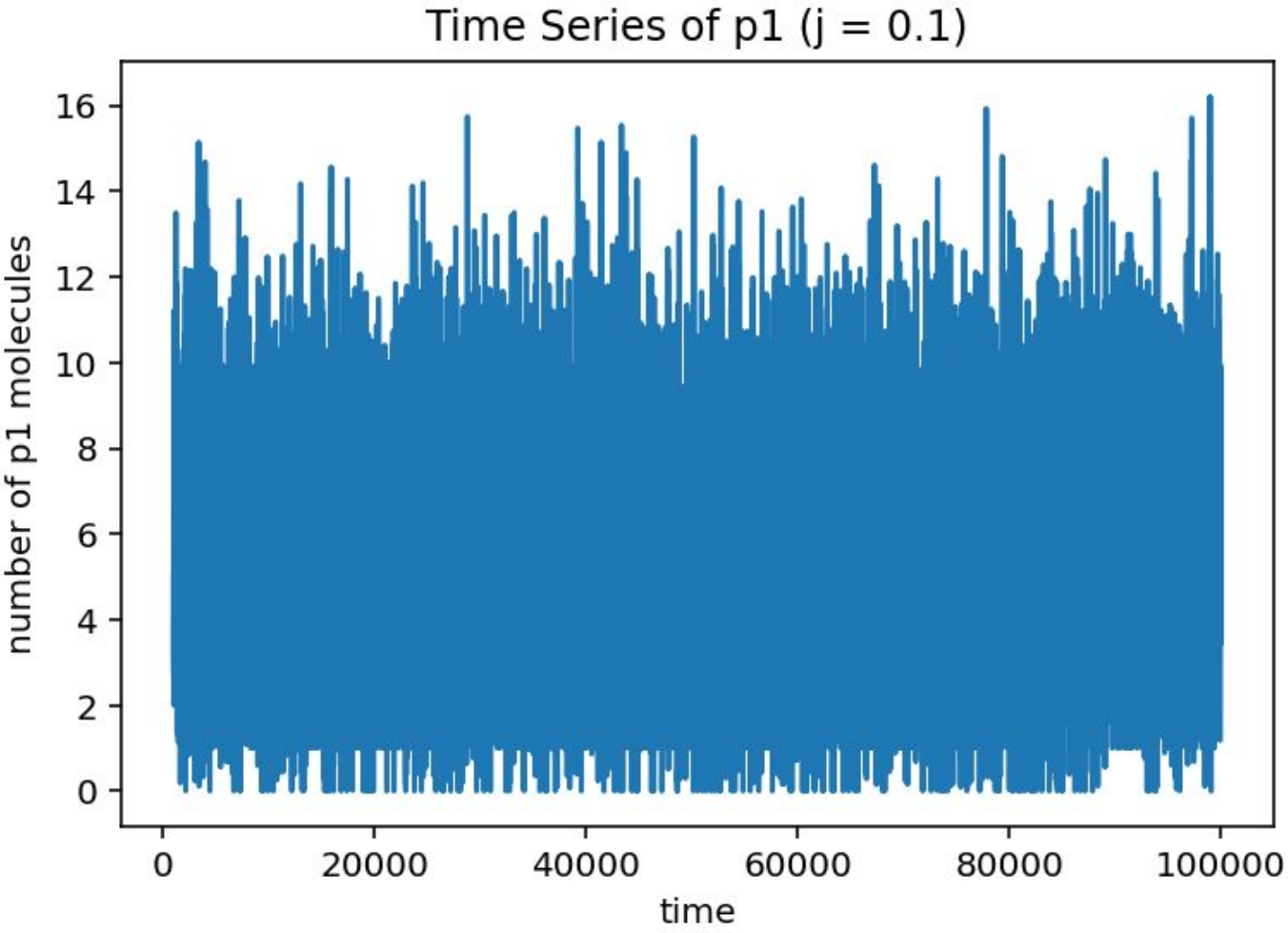
Thank you!

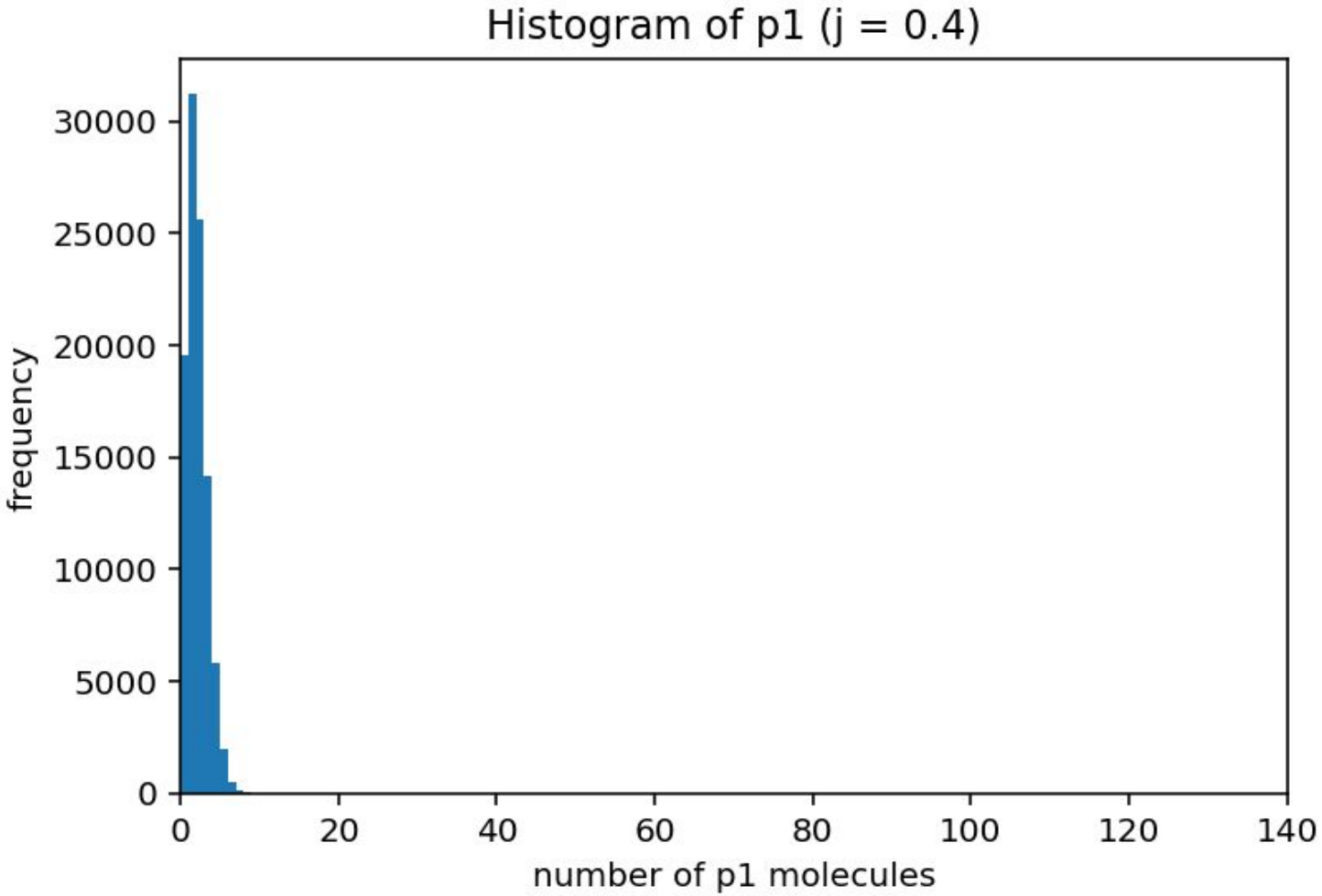

Time Series of p1 (j = 0.1)
number of p1 molecules
time
0
2
4
6
8
10
12
14
16
0
20000
40000
60000
80000
100000

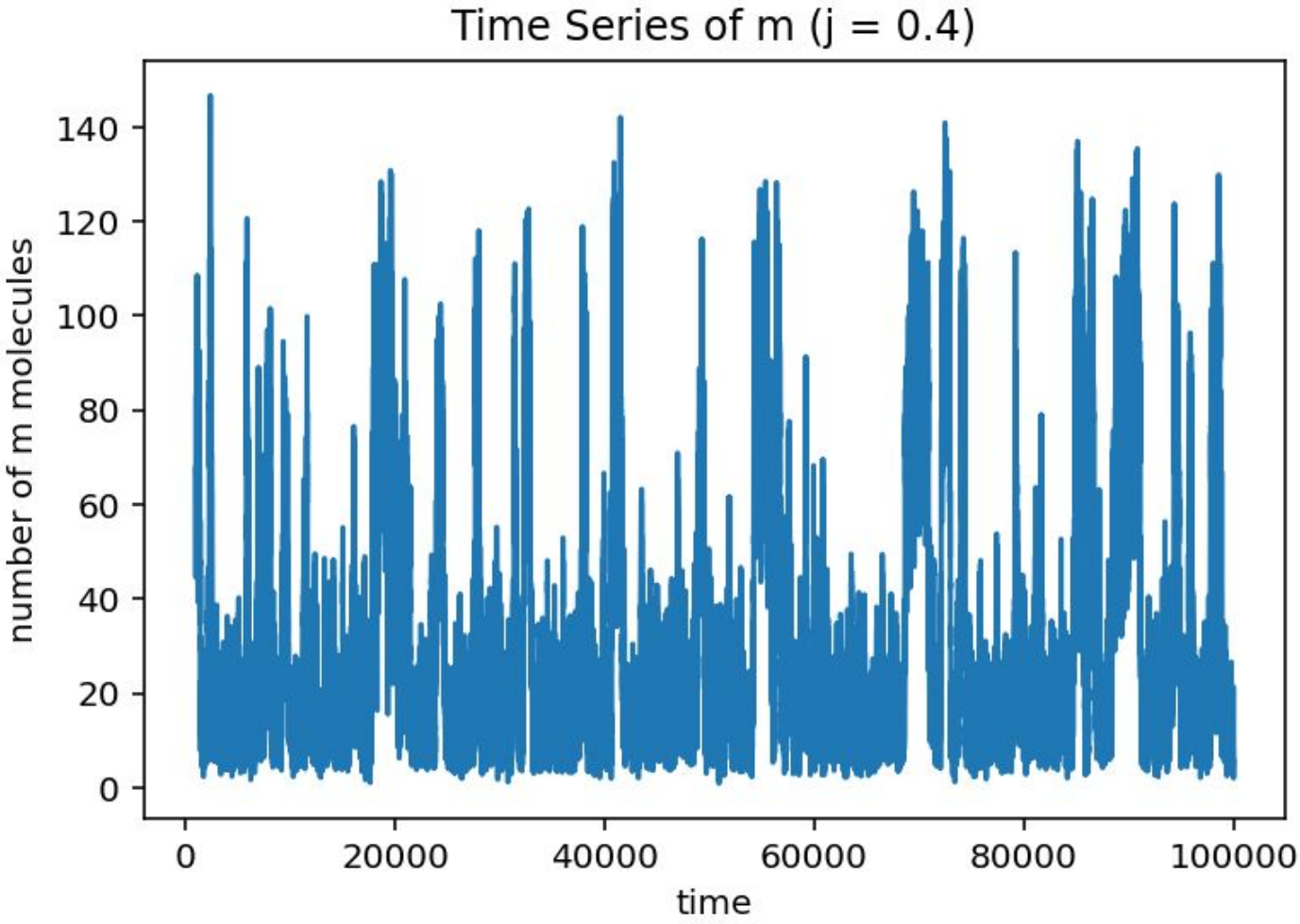

Histogram of p1 (j = 0.1)
frequency
17500
15000
12500
10000
7500
5000
2500
0
0
20
40
60
80
100
120
140
number of p1 molecules

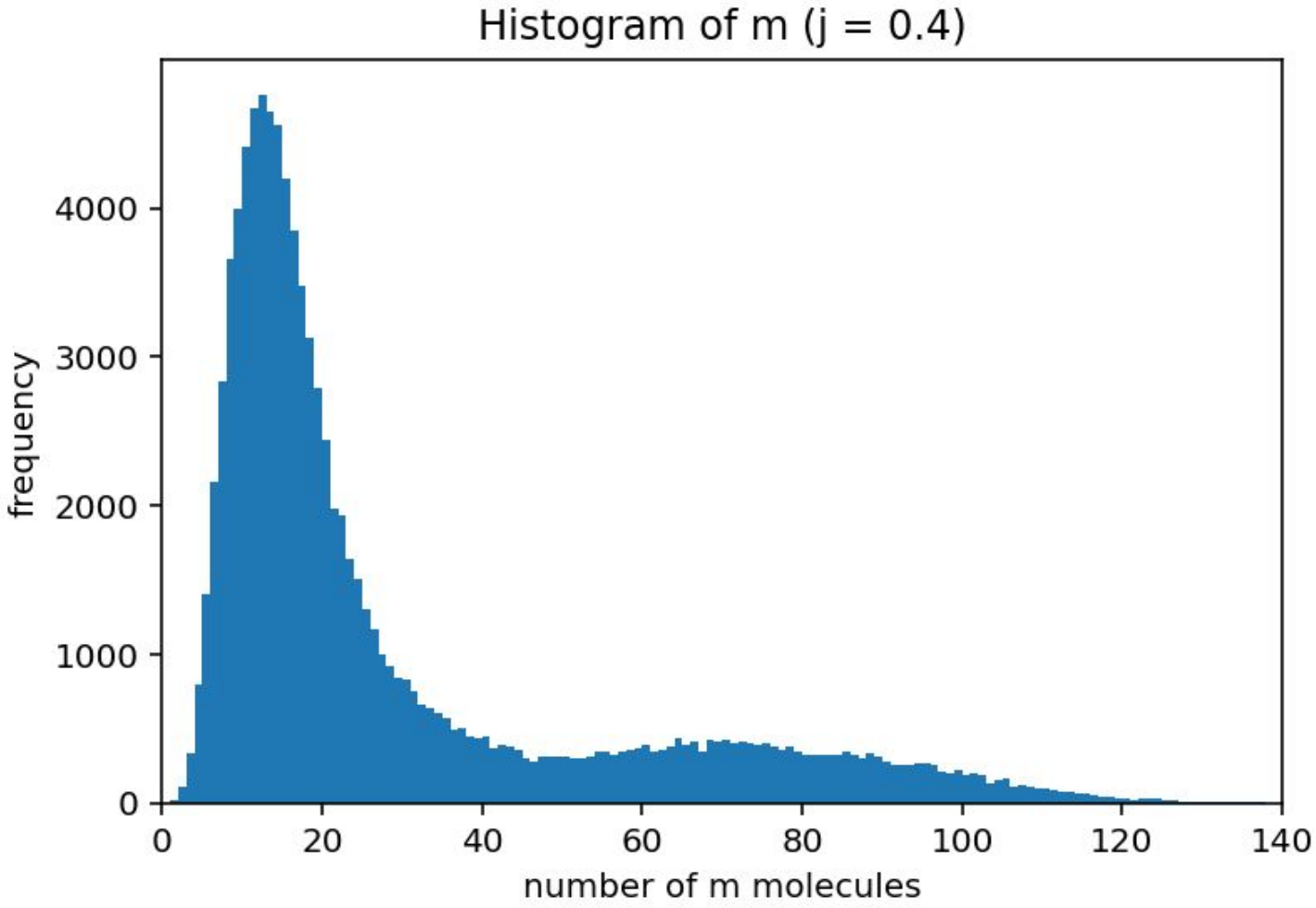

Time Series of m (j = 0.1)
number of m molecules
0
10
20
30
40
0
20000
40000
60000
80000
100000
time

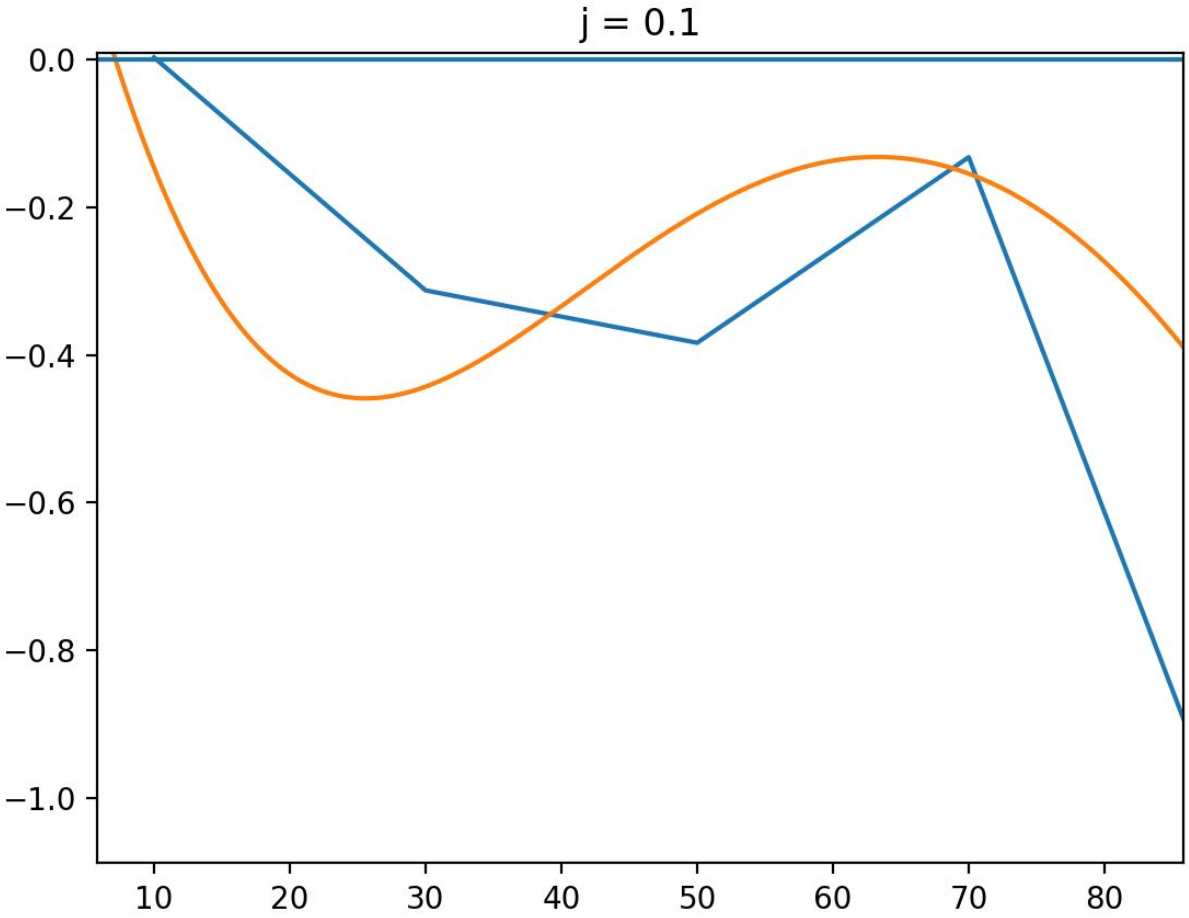

Histogram of m (j = 0.1)
frequency
10000
8000
6000
4000
2000
0
0
20
40
60
80
100
120
140
number of m molecules

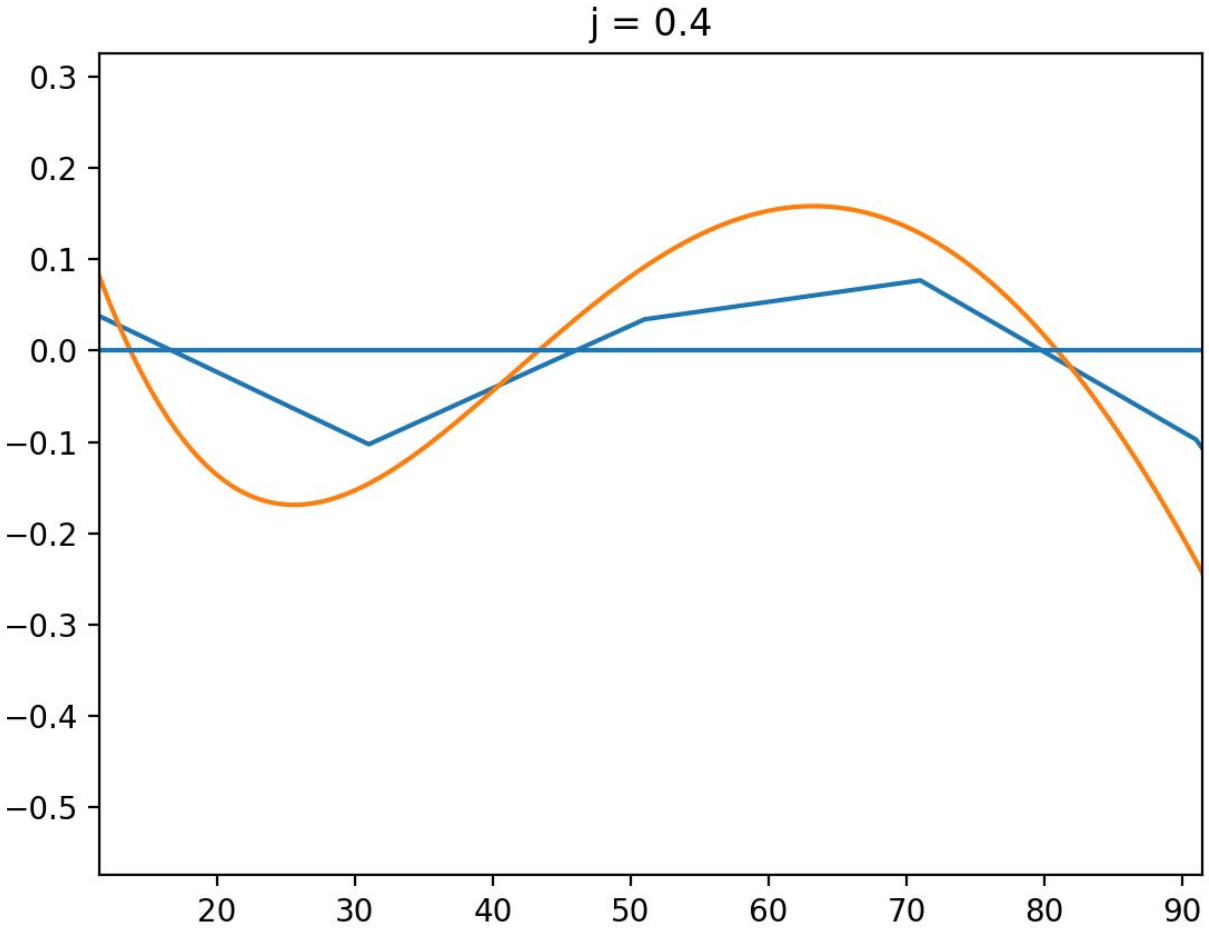

Time Series of p1 (j = 0.1)
number of p1 molecules
time
0
2
4
6
8
10
12
14
16
0
20000
40000
60000
80000
100000

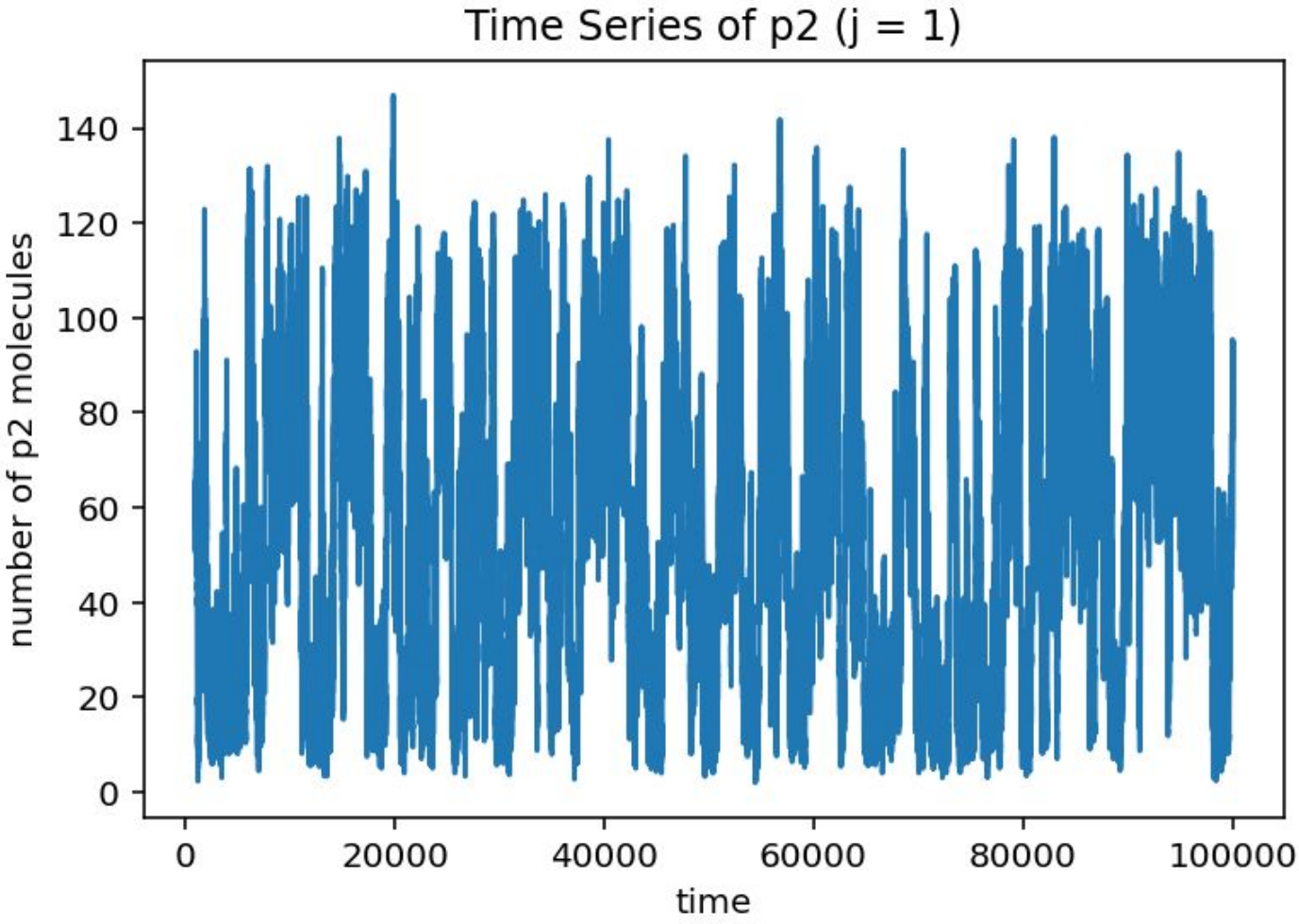

Histogram of p1 (j = 0.4)
frequency
30000
25000
20000
15000
10000
5000
0
0
20
40
60
80
100
120
140
number of p1 molecules

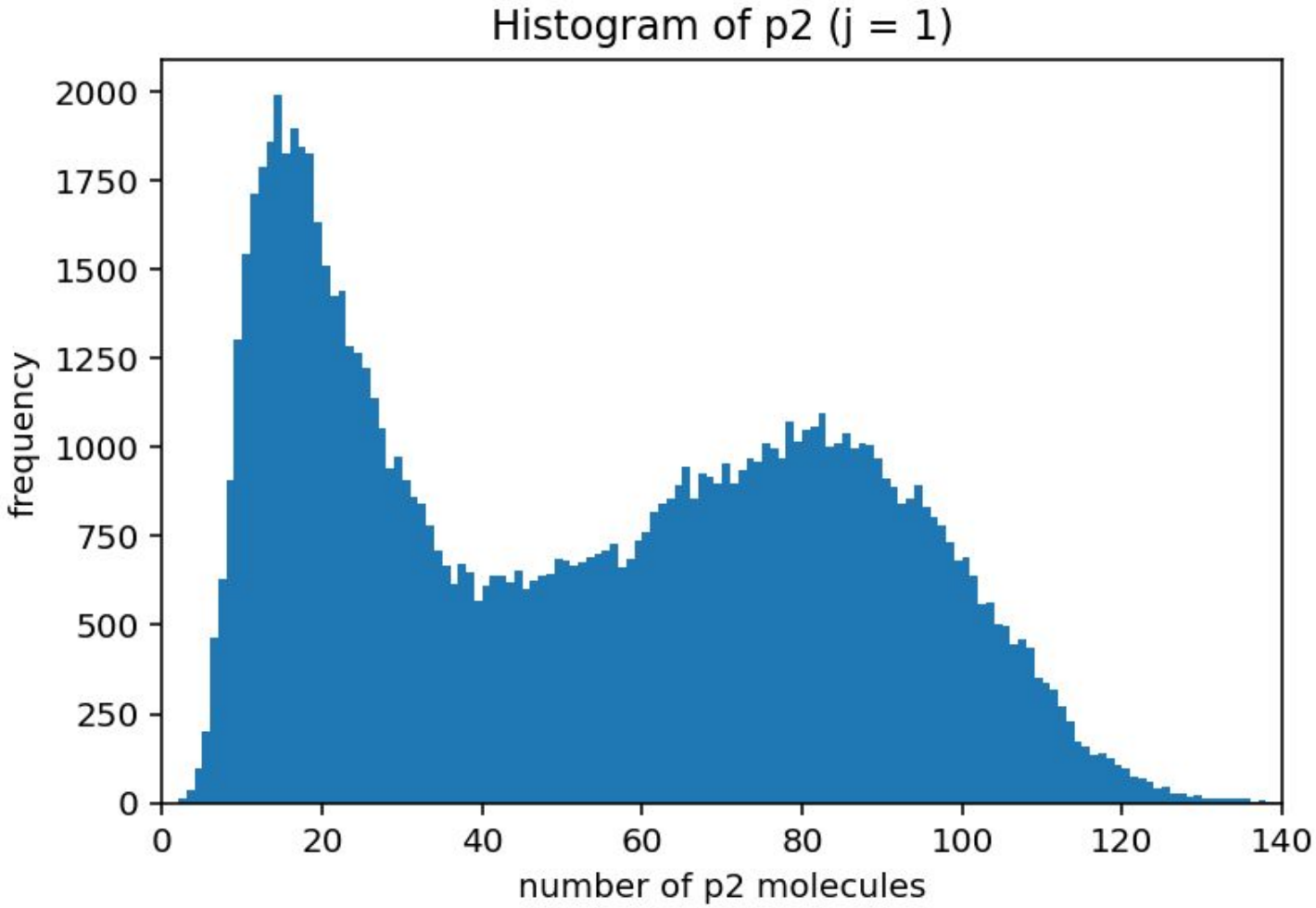

Time Series of m (j = 0.4)
number of m molecules
time
0
20
40
60
80
100
120
140
0
20000
40000
60000
80000
100000

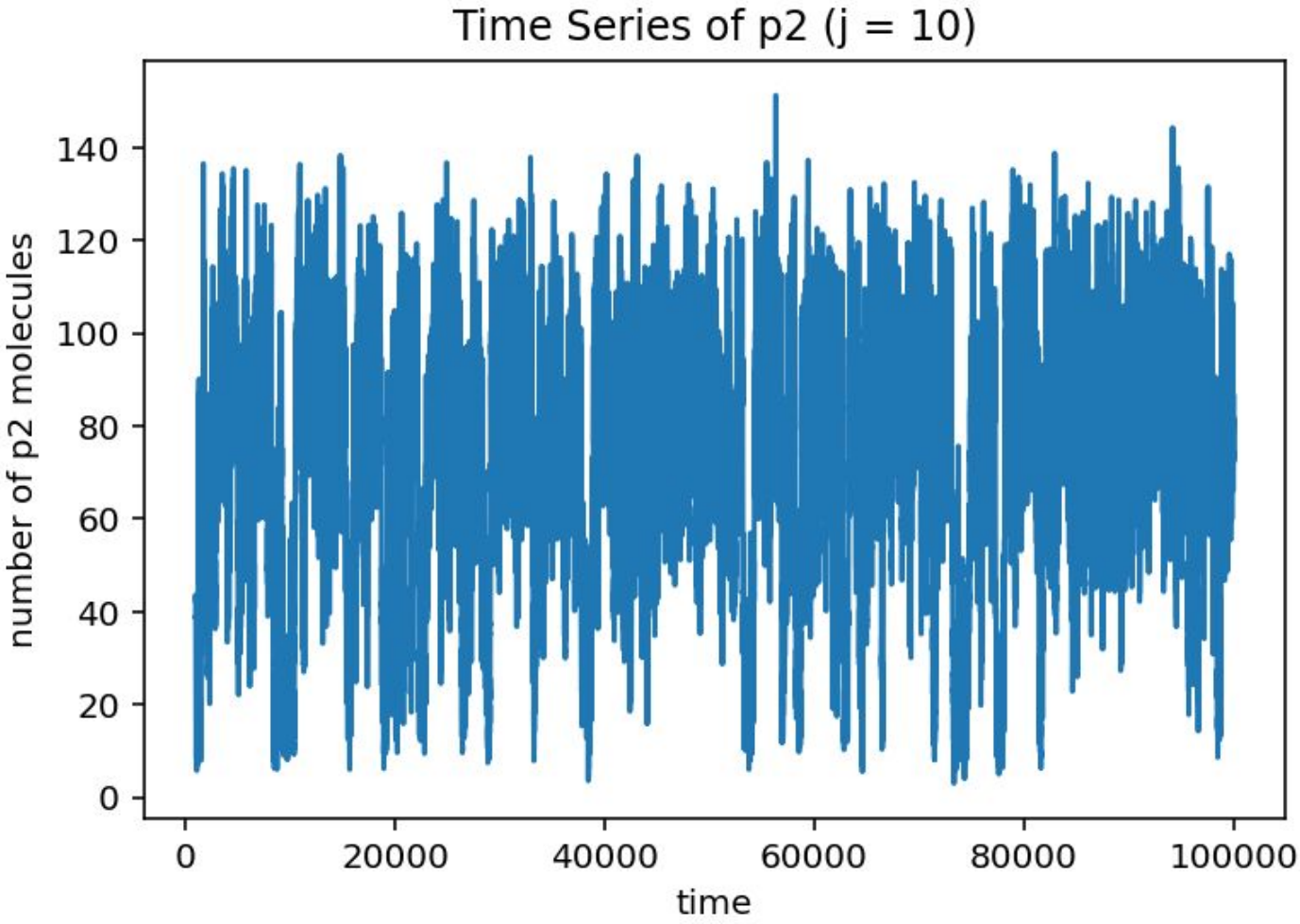

Histogram of m (j = 0.4)
frequency
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
number of m molecules

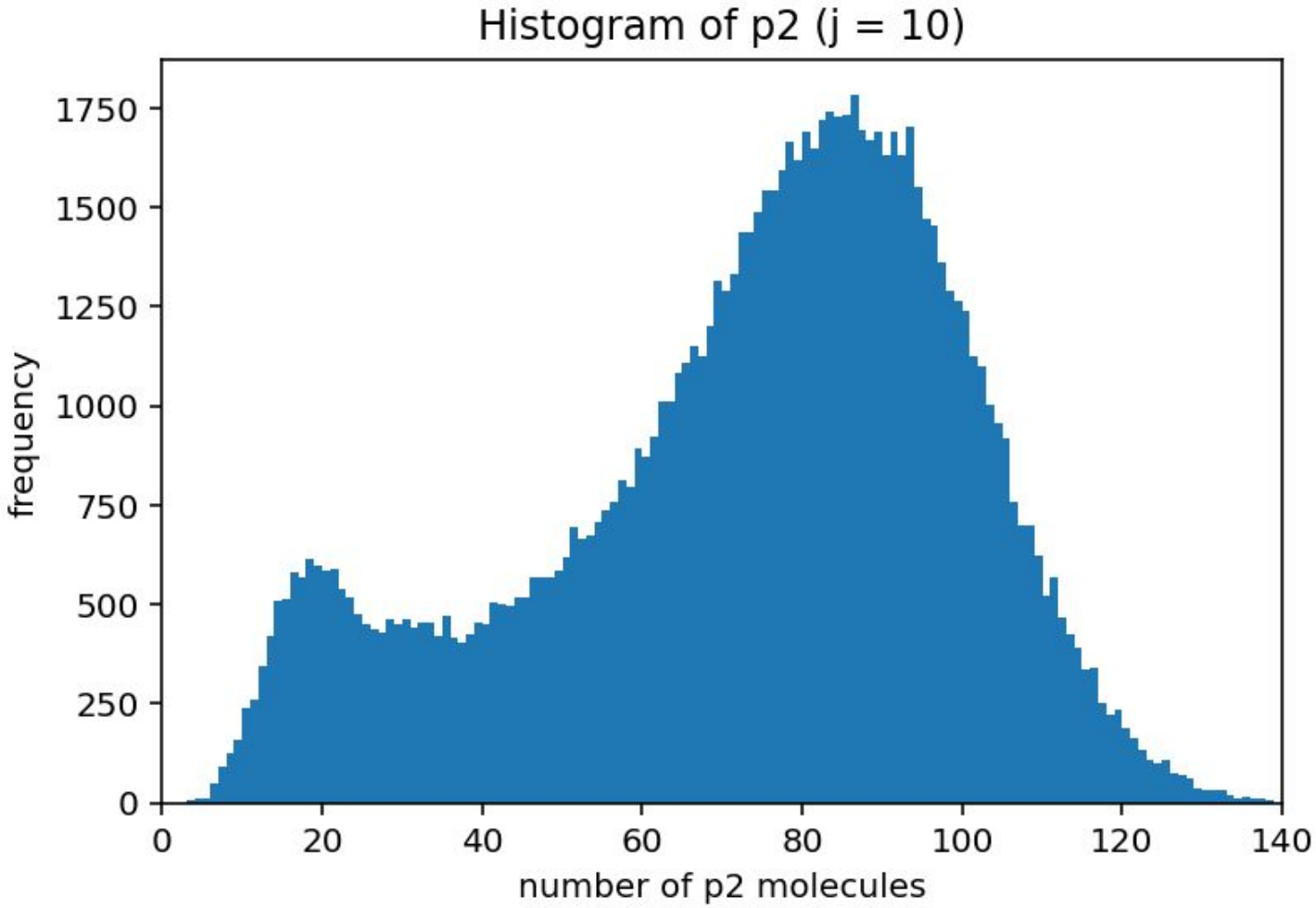

j = 0.1
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
10
20
30
40
50
60
70
80


j = 0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
20
30
40
50
60
70
80
90


Time Series of p2 (j = 1)
number of p2 molecules
time
0
20
40
60
80
100
120
140
0
20000
40000
60000
80000
100000


Histogram of p2 (j = 1)
frequency
number of p2 molecules
0
250
500
750
1000
1250
1500
1750
2000
0
20
40
60
80
100
120
140


Time Series of p2 (j = 10)
number of p2 molecules
0
20
40
60
80
100
120
140
0
20000
40000
60000
80000
100000
time


Histogram of p2 (j = 10)
frequency
number of p2 molecules
0
250
500
750
1000
1250
1500
1750
0
20
40
60
80
100
120
140